UNIT
9

SOLAR AND LUNAR ECLIPSES

Learning objectives:

- Understand how solar and lunar eclipses occur and their types.
- Identify the differences between solar and lunar eclipses.
- Recognize the safety measures to follow during an eclipse.

The Sun, Moon, and Earth create **eclipses** when one blocks the light of another. This chapter explains **solar and lunar eclipses**, how they happen, and why they are fascinating.

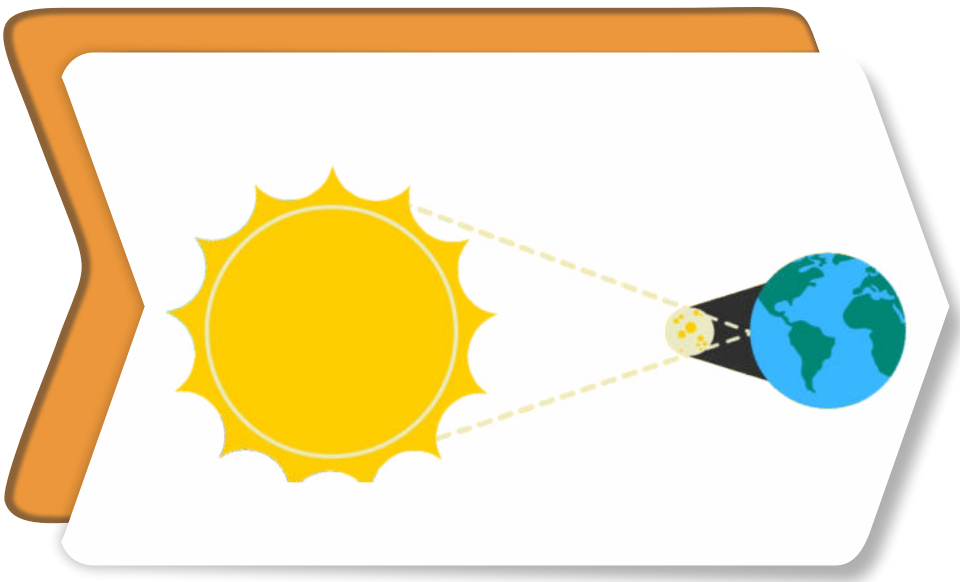
What Is a Solar Eclipse?

A **solar eclipse** occurs when the Moon comes in between the Sun and Earth, blocking sunlight from reaching the Earth. This makes the Sun appear dark for a short time. There are three types of solar eclipses:

1. Total Solar Eclipse: The Moon completely blocks the Sun.

2. Partial Solar Eclipse: The Moon blocks only part of the Sun.

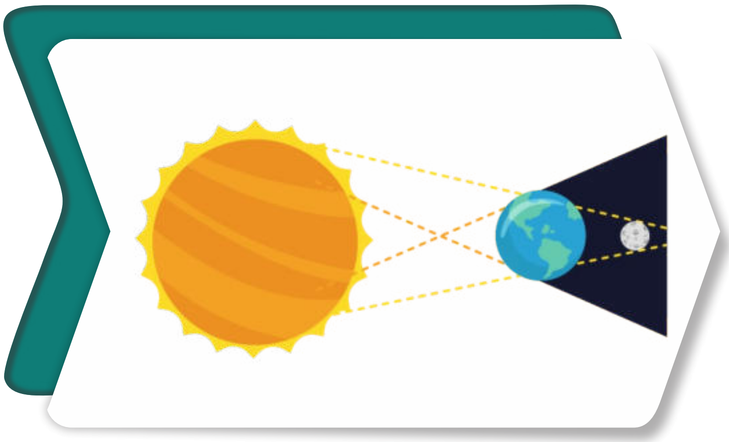
3. Annular Solar Eclipse: The Moon blocks the center of the Sun, leaving a bright ring around it.



What Is a Lunar Eclipse?

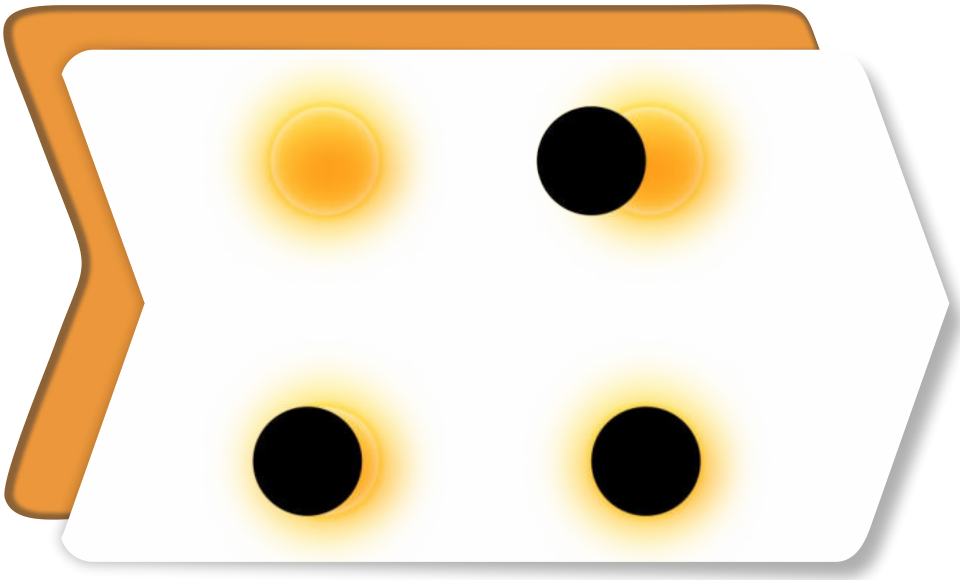
A **lunar eclipse** occurs when the Earth comes between the Sun and the Moon, blocking sunlight from reaching the Moon. The Moon appears dark or reddish. There are two main types of lunar eclipses:

1. **Total Lunar Eclipse**: The entire Moon moves into the Earth' shadow.
2. **Partial Lunar Eclipse**: Only part of the Moon enters the Earth's shadow.



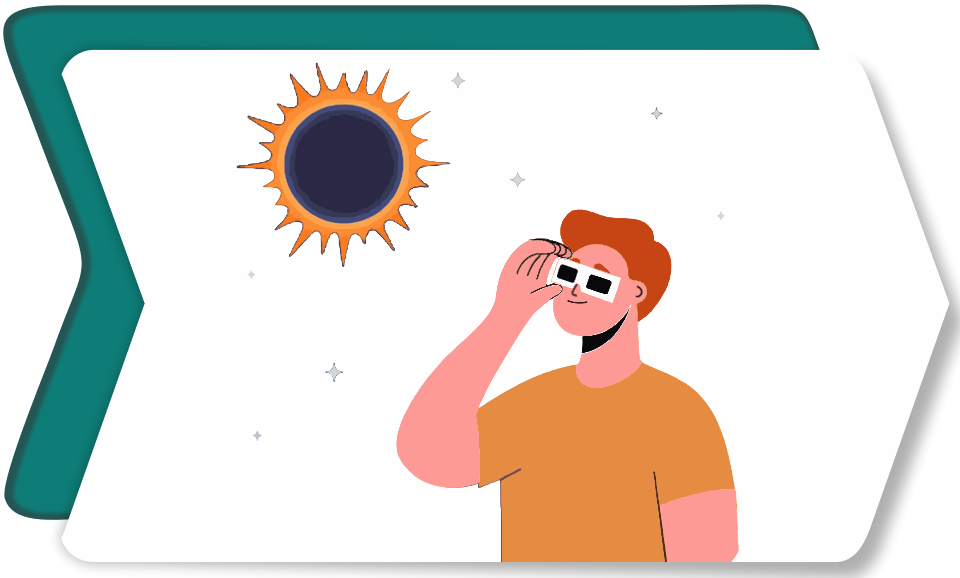
How Are Eclipses Different?

Solar Eclipse	Lunar Eclipse
Happens during the day	Happens at night
The Moon blocks sunlight from reaching the Earth	The Earth blocks sunlight from reaching the Moon
Can only be seen from certain parts of the Earth	Can be seen from anywhere on the night side of the Earth



Safety During Eclipses

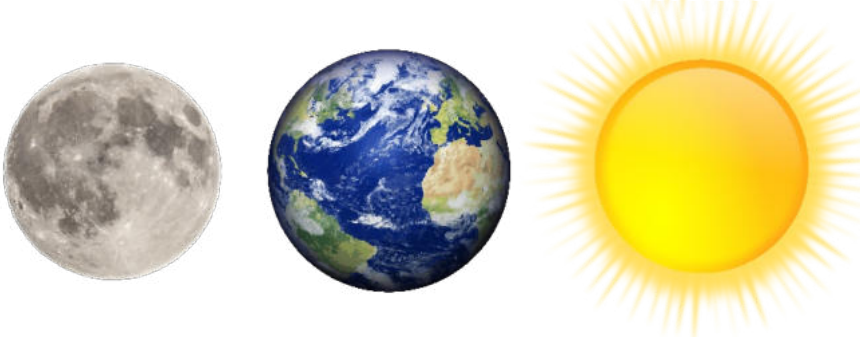
1. Never look directly at the Sun during a solar eclipse. Always use special eclipse glasses.
2. Lunar eclipses are safe to watch with the naked eye.
3. Use a pinhole projector to safely observe a solar eclipse.



Choose the Correct Answer.

Lunar Eclipse

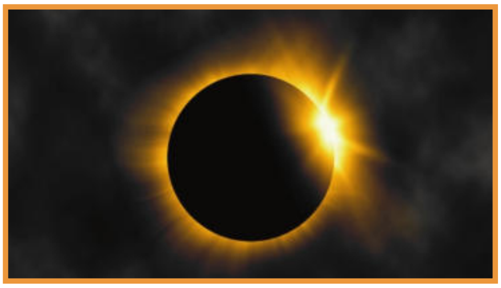
Solar Eclipse



Lunar Eclipse

Solar Eclipse

Write the Name of the Eclipse.



Fill in the blanks.

1. A solar eclipse occurs when the ______________ comes between the Sun and Earth.

2. During a lunar eclipse, the ______________ blocks sunlight from reaching the Moon.

3. ______________ eclipses happen during the day.

4. A total lunar eclipse makes the Moon appear ______________ in color.

5. ______________ eclipse is safe to watch without any special glasses.

6. During a solar eclipse, the Moon casts a ____________ on Earth.

7. A partial lunar eclipse happens when only a part of the Moon enters Earth's ____________.

Write "T" for True or "F" for False.

1. A lunar eclipse happens when the Moon blocks the Sun's light.

2. A solar eclipse occurs only at night.

3. You need special glasses to watch a solar eclipse safely.

4. The Earth blocks sunlight during a lunar eclipse.

5. The Moon turns reddish during a total lunar eclipse.

6. A solar eclipse can be seen from anywhere on Earth.

7.An annular eclipse shows a bright ring around the Moon.

8. Lunar eclipses are dangerous to observe without protection.

Choose the correct answer.

1. What happens during a solar eclipse?

- The Earth blocks sunlight from reaching the Moon
- The Moon blocks sunlight from reaching Earth
- The Sun blocks the Moon
- The Moon disappears

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2. Which eclipse occurs at night?

- Solar Eclipse
- Lunar Eclipse
- Both Solar and Lunar
- Neither Solar nor Lunar

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3. What is the Moon's colour during a total lunar eclipse?

- Yellow
- Blue
- Reddish
- Green

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4. Which object blocks sunlight during a lunar eclipse?

- The Moon
- The Sun
- The Earth
- The Stars

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5. Why is it unsafe to look at a solar eclipse directly?

- The Sun becomes too bright
- The Moon blocks the light
- It can damage your eyes
- The Sun disappears

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6. When can a lunar eclipse be observed?

- Only during the day
- Only during a new moon
- Only at night
- Only during a solar eclipse

Answer the following questions.

Q1. Differentiate between solar and lunar eclipse?

Q2. Why is it dangerous to look directly at a solar eclipse?

Q3. How is a total solar eclipse different from an annular solar eclipse?

Q4. Why does the Moon appear reddish during a total lunar eclipse?

Q5. What happens during a partial lunar eclipse?

Q6. How can we safely observe a solar eclipse?

Q 7.Why can lunar eclipses be seen from more places on Earth compared to solar eclipses?

Critical Thinking

Why do you think we don't see solar and lunar eclipses every month, even though the Sun, Moon, and Earth are always moving?

Create a Model of an Eclipse

ACTIVITY

Materials Needed:

A flashlight, A small ball (Moon) and a larger ball (Earth).

Steps:

1. Place the flashlight on a table to act as the Sun.

2. Hold the large ball (Earth) in front of the flashlight.

3. Put the smaller ball (Moon) behind the Earth to create a lunar eclipse. Observe the shadow on the Moon.

4. Move the smaller ball between the flashlight and the Earth to create a solar eclipse. Observe the shadow on Earth.